



```
In [1]: pip install pandas
```

```
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: pandas in c:\users\hp\appdata\roaming\python\python313\site-packages (2.3.3)
Requirement already satisfied: numpy>=1.26.0 in c:\users\hp\appdata\roaming\python\python313\site-packages (from pandas) (2.3.4)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\hp\appdata\roaming\python\python313\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\hp\appdata\roaming\python\python313\site-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in c:\users\hp\appdata\roaming\python\python313\site-packages (from pandas) (2025.2)
Requirement already satisfied: six>=1.5 in c:\users\hp\appdata\roaming\python\python313\site-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
Note: you may need to restart the kernel to use updated packages.
```

Pandas

```
In [2]: import pandas as pd
```

```
In [3]: df = pd.read_csv(r"c:\Users\hp\Documents\indian_sales_performance_dataset.csv")
```

```
In [4]: pd
```

```
Out[4]: <module 'pandas' from 'C:\\Users\\hp\\AppData\\Roaming\\Python\\Python313\\site-packages\\pandas\\__init__.py'>
```

```
In [5]: df
```

Out[5]:

	Employee_ID	Employee_Name	Gender	Age	Department	Region	Exp
0	3001	Aarohi	Female	36	Sales	Assam	
1	3002	Ananya	Female	35	IT	Bihar	
2	3003	Ishita	Female	45	Finance	Uttar Pradesh	
3	3004	Vihaan	Male	56	Marketing	Uttar Pradesh	
4	3005	Dhruv	Male	31	Finance	Punjab	
...	...	...	...	...	...	...	...
195	3196	Aniket	Male	26	Finance	Uttarakhand	
196	3197	Dhruv	Male	23	HR	Odisha	
197	3198	Aryan	Male	36	Finance	Assam	
198	3199	Karthik	Male	27	Finance	Assam	
199	3200	Sneha	Female	34	HR	West Bengal	

200 rows × 10 columns

In [6]: `df.head()`

Out[6]:

	Employee_ID	Employee_Name	Gender	Age	Department	Region	Experienc
0	3001	Aarohi	Female	36	Sales	Assam	
1	3002	Ananya	Female	35	IT	Bihar	
2	3003	Ishita	Female	45	Finance	Uttar Pradesh	
3	3004	Vihaan	Male	56	Marketing	Uttar Pradesh	
4	3005	Dhruv	Male	31	Finance	Punjab	

In [7]: `df.tail()`

Out[7]:

	Employee_ID	Employee_Name	Gender	Age	Department	Region	Exp
195	3196	Aniket	Male	26	Finance	Uttarakhand	
196	3197	Dhruv	Male	23	HR	Odisha	
197	3198	Aryan	Male	36	Finance	Assam	
198	3199	Karthik	Male	27	Finance	Assam	
199	3200	Sneha	Female	34	HR	West Bengal	

```
In [8]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 10 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Employee_ID           200 non-null    int64
1   Employee_Name         200 non-null    object
2   Gender                200 non-null    object
3   Age                   200 non-null    int64
4   Department            200 non-null    object
5   Region                200 non-null    object
6   Experience_Years       200 non-null    int64
7   Monthly_Sales         200 non-null    int64
8   Customer_Satisfaction 200 non-null    int64
9   Performance_Rating    200 non-null    object
dtypes: int64(5), object(5)
memory usage: 15.8+ KB
```

```
In [9]: df.describe()
```

```
Out[9]:
```

	Employee_ID	Age	Experience_Years	Monthly_Sales	Customer_Sati
count	200.000000	200.00000	200.000000	200.000000	200.000000
mean	3100.500000	38.59500	14.920000	81435.635000	81.435635
std	57.879185	10.82529	8.673508	35862.006242	35.862006
min	3001.000000	21.00000	0.000000	25155.000000	25.155000
25%	3050.750000	29.00000	7.000000	51016.500000	51.016500
50%	3100.500000	38.00000	16.000000	80172.000000	80.172000
75%	3150.250000	48.00000	23.000000	110201.250000	110.201250
max	3200.000000	57.00000	29.000000	149661.000000	149.661000

```
In [10]: df.shape
```

```
Out[10]: (200, 10)
```

```
In [11]: df.columns
```

```
Out[11]: Index(['Employee_ID', 'Employee_Name', 'Gender', 'Age', 'Department', 'Region',
                'Experience_Years', 'Monthly_Sales', 'Customer_Satisfaction',
                'Performance_Rating'],
              dtype='object')
```

```
In [12]: df.values
```

```
Out[12]: array([[3001, 'Aarohi', 'Female', ..., 48648, 2, 'Average'],
                [3002, 'Ananya', 'Female', ..., 75729, 3, 'Good'],
                [3003, 'Ishita', 'Female', ..., 66971, 1, 'Average'],
                ...,
                [3198, 'Aryan', 'Male', ..., 75939, 10, 'Good'],
                [3199, 'Karthik', 'Male', ..., 54516, 2, 'Excellent'],
                [3200, 'Sneha', 'Female', ..., 81526, 3, 'Excellent']],
          shape=(200, 10), dtype=object)
```

```
In [13]: df.dtypes
```

```
Out[13]: Employee_ID      int64
Employee_Name      object
Gender              object
Age                int64
Department          object
Region              object
Experience_Years    int64
Monthly_Sales       int64
Customer_Satisfaction int64
Performance_Rating  object
dtype: object
```

```
In [14]: df.loc[0]
```

```
Out[14]: Employee_ID      3001
Employee_Name      Aarohi
Gender              Female
Age                36
Department          Sales
Region              Assam
Experience_Years    2
Monthly_Sales       48648
Customer_Satisfaction 2
Performance_Rating  Average
Name: 0, dtype: object
```

```
In [15]: df.iloc[0,1]
```

```
Out[15]: 'Aarohi'
```

```
In [16]: filtered_df = df.query('Age > 25')
filtered_df
```

Out[16]:

	Employee_ID	Employee_Name	Gender	Age	Department	Region	Exp
0	3001	Aarohi	Female	36	Sales	Assam	
1	3002	Ananya	Female	35	IT	Bihar	
2	3003	Ishita	Female	45	Finance	Uttar Pradesh	
3	3004	Vihaan	Male	56	Marketing	Uttar Pradesh	
4	3005	Dhruv	Male	31	Finance	Punjab	
...	...	...	...	...	...	...	...
194	3195	Meera	Female	34	HR	Rajasthan	
195	3196	Aniket	Male	26	Finance	Uttarakhand	
197	3198	Aryan	Male	36	Finance	Assam	
198	3199	Karthik	Male	27	Finance	Assam	
199	3200	Sneha	Female	34	HR	West Bengal	

171 rows × 10 columns

In [17]:

```
df_dropped = df.drop(columns=['Age'])
df_dropped
```

Out[17]:

	Employee_ID	Employee_Name	Gender	Department	Region	Experience
0	3001	Aarohi	Female	Sales	Assam	
1	3002	Ananya	Female	IT	Bihar	
2	3003	Ishita	Female	Finance	Uttar Pradesh	
3	3004	Vihaan	Male	Marketing	Uttar Pradesh	
4	3005	Dhruv	Male	Finance	Punjab	
...	...	...	...	...	...	...
195	3196	Aniket	Male	Finance	Uttarakhand	
196	3197	Dhruv	Male	HR	Odisha	
197	3198	Aryan	Male	Finance	Assam	
198	3199	Karthik	Male	Finance	Assam	
199	3200	Sneha	Female	HR	West Bengal	

200 rows × 9 columns

```
In [18]: df_renamed = df.rename(columns={'Age':'Life Line'})
df_renamed
```

```
Out[18]:
```

	Employee_ID	Employee_Name	Gender	Life Line	Department	Region	Exp
0	3001	Aarohi	Female	36	Sales	Assam	
1	3002	Ananya	Female	35	IT	Bihar	
2	3003	Ishita	Female	45	Finance	Uttar Pradesh	
3	3004	Vihaan	Male	56	Marketing	Uttar Pradesh	
4	3005	Dhruv	Male	31	Finance	Punjab	
...	...	...	...	...	...	...	...
195	3196	Aniket	Male	26	Finance	Uttarakhand	
196	3197	Dhruv	Male	23	HR	Odisha	
197	3198	Aryan	Male	36	Finance	Assam	
198	3199	Karthik	Male	27	Finance	Assam	
199	3200	Sneha	Female	34	HR	West Bengal	

200 rows × 10 columns

```
In [19]: df_sorted = df.sort_values(by='Age')
df_sorted
```

```
Out[19]:
```

	Employee_ID	Employee_Name	Gender	Age	Department	Region	Exp
40	3041	Aarav	Male	21	HR	Telangana	
153	3154	Aryan	Male	21	Sales	Assam	
175	3176	Meera	Female	21	IT	Chhattisgarh	
23	3024	Ira	Female	22	Operations	Jharkhand	
111	3112	Ishaan	Male	22	IT	Uttarakhand	
...	...	...	...	...	...	...	...
100	3101	Sai	Male	56	IT	Telangana	
101	3102	Ira	Female	57	IT	Madhya Pradesh	
76	3077	Meera	Female	57	Finance	Maharashtra	
57	3058	Siddharth	Male	57	IT	Telangana	
187	3188	Aditi	Female	57	Finance	West Bengal	

200 rows × 10 columns

```
In [20]: df_filled = df.fillna(0)
df_filled
```

```
Out[20]:
```

	Employee_ID	Employee_Name	Gender	Age	Department	Region	Exp
0	3001	Aarohi	Female	36	Sales	Assam	
1	3002	Ananya	Female	35	IT	Bihar	
2	3003	Ishita	Female	45	Finance	Uttar Pradesh	
3	3004	Vihaan	Male	56	Marketing	Uttar Pradesh	
4	3005	Dhruv	Male	31	Finance	Punjab	
...	...	...	...	...	...	...	...
195	3196	Aniket	Male	26	Finance	Uttarakhand	
196	3197	Dhruv	Male	23	HR	Odisha	
197	3198	Aryan	Male	36	Finance	Assam	
198	3199	Karthik	Male	27	Finance	Assam	
199	3200	Sneha	Female	34	HR	West Bengal	

200 rows × 10 columns

```
In [21]: df_unique = df.drop_duplicates()
```

df_unique

Out[21]:

	Employee_ID	Employee_Name	Gender	Age	Department	Region	Exp
0	3001	Aarohi	Female	36	Sales	Assam	
1	3002	Ananya	Female	35	IT	Bihar	
2	3003	Ishita	Female	45	Finance	Uttar Pradesh	
3	3004	Vihaan	Male	56	Marketing	Uttar Pradesh	
4	3005	Dhruv	Male	31	Finance	Punjab	
...	...	...	...	...	...	...	...
195	3196	Aniket	Male	26	Finance	Uttarakhand	
196	3197	Dhruv	Male	23	HR	Odisha	
197	3198	Aryan	Male	36	Finance	Assam	
198	3199	Karthik	Male	27	Finance	Assam	
199	3200	Sneha	Female	34	HR	West Bengal	

200 rows × 10 columns

```
In [33]: df_replaced = df.replace({'Aarohi': 'Kavya'})
df_replaced
```

Out[33]:	Employee_ID	Employee_Name	Gender	Age	Department	Region	Exp
0	3001	Kavya	Female	36	Sales	Assam	
1	3002	Ananya	Female	35	IT	Bihar	
2	3003	Ishita	Female	45	Finance	Uttar Pradesh	
3	3004	Vihaan	Male	56	Marketing	Uttar Pradesh	
4	3005	Dhruv	Male	31	Finance	Punjab	
...	...	...	...	...	...	...	...
195	3196	Aniket	Male	26	Finance	Uttarakhand	
196	3197	Dhruv	Male	23	HR	Odisha	
197	3198	Aryan	Male	36	Finance	Assam	
198	3199	Karthik	Male	27	Finance	Assam	
199	3200	Sneha	Female	34	HR	West Bengal	

200 rows × 10 columns

```
In [23]: grouped_df = df.groupby('Employee_Name').sum()
grouped_df
```

Out[23]:

Employee_ID			Gender				
Employee_Name							
Aanya	18664		Female	Female	Female	Female	Female
Aarav	21647			Male	Male	Male	Male
Aarohi	18519		Female	Female	Female	Female	Female
Abhishek	9350						Male
Aditi	24888	Female	Female	Female	Female	Female	Female
Aditya	9263						Male
Ananya	12290				Female	Female	Female
Aniket	18703				Male	Male	Male
Arjun	6334						Male
Aryan	25016		Male	Male	Male	Male	Male
Dhruv	21567			Male	Male	Male	Male
Diya	18740		Female	Female	Female	Female	Female
Harsh	6132						Male
Ira	18383		Female	Female	Female	Female	Female
Ishaan	12443					Male	Male
Ishita	9133					Female	Female
Karthik	9486						Male
Kavya	9260					Female	Female
Krishna	6176						Male
Manish	9439						Male
Meera	30866	Female	Female	Female	Female	Female	Female
Mira	18805		Female	Female	Female	Female	Female
Neha	6308						Female
Nikhil	6111						Male
Nisha	15422			Female	Female	Female	Female
Pooja	18527		Female	Female	Female	Female	Female
Priya	24900	Female	Female	Female	Female	Female	Female
Rahul	18941					Male	Male
Reyansh	27694		Male	Male	Male	Male	Male

Employee_ID		Gende					
Employee_Name							
Riya	18741	Female	Female	Female	Female	Female	Femal
Rohan	12371					Male	Male
Saanvi	9378					Female	Female
Sai	21656			Male	Male	Male	Male
Shreya	18391	Female	Female	Female	Female	Female	Femal
Siddharth	15428					Male	Male
Simran	18630	Female	Female	Female	Female	Female	Femal
Sneha	12438				Female	Female	Female
Tanvi	15499	Female	Female	Female	Female	Female	Femal
Vihaan	6042						Male
Vivaan	18519				Male	Male	Male

```
In [24]: agg_df = df.groupby('Employee_Name').agg({'Age': 'mean'})
agg_df
```

Out[24]:

Age

Employee_Name	
Aanya	38.666667
Aarav	35.285714
Aarohi	37.833333
Abhishek	31.000000
Aditi	41.250000
Aditya	35.333333
Ananya	36.000000
Aniket	33.500000
Arjun	35.000000
Aryan	42.000000
Dhruv	33.142857
Diya	38.833333
Harsh	44.500000
Ira	39.500000
Ishaan	37.250000
Ishita	31.666667
Karthik	40.333333
Kavya	32.666667
Krishna	43.500000
Manish	40.000000
Meera	36.400000
Mira	48.833333
Neha	46.000000
Nikhil	36.000000
Nisha	39.800000
Pooja	47.500000
Priya	35.625000
Rahul	32.333333
Reyansh	44.777778
Riya	41.000000

Age	
Employee_Name	
Rohan	39.250000
Saanvi	36.666667
Sai	39.142857
Shreya	43.833333
Siddharth	36.000000
Simran	39.833333
Sneha	35.750000
Tanvi	31.000000
Vihaan	48.000000
Vivaan	37.000000

```
In [25]: agg_df = df.groupby('Employee_Name').agg({'Monthly_Sales': 'mean'})
agg_df
```

Out[25]:

Monthly_Sales

Employee_Name	
Aanya	72642.833333
Aarav	77352.571429
Aarohi	59824.500000
Abhishek	65031.333333
Aditi	76594.875000
Aditya	68760.000000
Ananya	61770.250000
Aniket	87175.833333
Arjun	71212.500000
Aryan	66579.750000
Dhruv	75664.571429
Diya	89312.166667
Harsh	65898.500000
Ira	104224.166667
Ishaan	89559.000000
Ishita	50253.666667
Karthik	68509.000000
Kavya	89778.666667
Krishna	26820.000000
Manish	87004.000000
Meera	99371.400000
Mira	113985.500000
Neha	83259.000000
Nikhil	59758.500000
Nisha	76787.800000
Pooja	82880.333333
Priya	107478.375000
Rahul	80745.166667
Reyansh	85422.666667
Riya	91168.166667

Monthly_Sales	
Employee_Name	
Rohan	73566.250000
Saanvi	106765.333333
Sai	93496.714286
Shreya	83574.333333
Siddharth	57060.400000
Simran	71807.166667
Sneha	40647.250000
Tanvi	98551.400000
Vihaan	77051.500000
Vivaan	94878.666667

```
In [26]: count_df = df.groupby('Monthly_Sales').count()
count_df
```

```
Out[26]:
```

Employee_ID	Employee_Name	Gender	Age	Department	Regio
Monthly_Sales					
25155	1	1	1	1	1
25282	1	1	1	1	1
25587	1	1	1	1	1
25645	1	1	1	1	1
26089	1	1	1	1	1
...	...	...	...	...	...
147046	1	1	1	1	1
147232	1	1	1	1	1
147775	1	1	1	1	1
148950	1	1	1	1	1
149661	1	1	1	1	1

200 rows × 9 columns

```
In [27]: sum_df = df.groupby('Age').sum()
sum_df
```

Out[27]:

	Employee_ID	Employee_Name	
Age			
21	9371	AaravAryanMeera	
22	21817	IraDhruvIshaanRahulRahulAarohiMeera	
23	21960	IshitaVivaanAnanyaAaravSaiSaanviDhruv	
24	24534	MeeraSiddharthAaravPriyaAditiAniketTanviDiya	Female
25	12469	AdityaSiddharthRahulAbhishek	
26	15768	AniketAbhishekAanyaPriyaAniket	
27	18478	AditiIshitaRohanTanviShreyaKarthik	
28	24851	KavyaDhruvSnehaIshaanMeeraRiyaSaiAditi	Female
29	15462	AanyaSnehaKavyaAanyaVivaan	
30	15299	PoojaAditiAniketNikhilTanvi	
31	15390	DhruvNishaDiyaTanviRiya	
32	12639	DiyaArjunSimranPriya	
33	21443	ReyanshIraSaanviReyanshAryanHarshNisha	
34	15661	RiyaSiddharthPriyaMeeraSneha	
35	15322	AnanyaMeeraReyanshPriyaRahul	
36	28007	AarohiSaiSimranRohanAnanyaManishVivaanSimranAryan	FemaleM
37	9343	SimranIraSai	
38	18713	AdityaVivaanShreyaKrishnaMiraArjun	
39	12235	SaiDhruvRohanAarohi	
40	12494	VihaanNehaNishaSiddharth	
41	21662	VivaanAarohiAaravNishaDhruvKavyaManish	
42	27958	ReyanshNikhilAryanAbhishekAanyaAaravPriyaDiyaR...	MaleM
43	15752	ManishTanviKarthikAdityaAarohi	
44	15455	IraIraMiraPoojaRahul	
45	12183	IshitaPriyaMeeraAniket	
46	15410	AarohiShreyaSimranAaravRahul	
47	12360	MeeraPriyaIshaanPooja	
48	12294	ShreyaDhruvShreyaRiya	
49	6152	KrishnaRiya	

	Employee_ID	Employee_Name	
Age			
50	25010	DiyaAnanyaMiraAryanAniketAaravAryanMira	Female
51	9204	AryanMeeraKarthik	
52	12459	SnehalshaanSimranNeha	
53	18685	AanyaReyanshAditiPoojaAanyaAryan	
54	12378	NishaReyanshDiyaSaanvi	
55	21760	VivaanMiraPoojaSaiRohanReyanshAditi	
56	27697	VihaanHarshPoojaShreyaReyanshSaiAditiRiyaMira	MaleMaleFemale
57	12425	SiddharthMeeraIraAditi	

```
In [28]: max_df = df.groupby('Monthly_Sales').max()
max_df
```

```
Out[28]:
```

	Employee_ID	Employee_Name	Gender	Age	Department	Region
Monthly_Sales						
25155	3164	Simran	Female	36	Marketing	Maharashtra
25282	3046	Dhruv	Male	48	HR	West Bengal
25587	3122	Abhishek	Male	26	IT	Punjab
25645	3107	Sneha	Female	28	Operations	Haryana
26089	3160	Krishna	Male	38	HR	Chhattisgarh
...	...	...	...	...	...	...
147046	3183	Priya	Female	26	Sales	Maharashtra
147232	3110	Tanvi	Female	24	Operations	West Bengal
147775	3020	Sai	Male	36	Operations	Chhattisgarh
148950	3165	Simran	Female	32	Marketing	Jharkhand
149661	3065	Vivaan	Male	38	Finance	Chhattisgarh

200 rows × 9 columns

```
In [29]: min_df = df.groupby('Monthly_Sales').min()
min_df
```

Out[29]:

	Employee_ID	Employee_Name	Gender	Age	Department	R
Monthly_Sales						
	25155	3164	Simran	Female	36	Marketing
	25282	3046	Dhruv	Male	48	HR
	25587	3122	Abhishek	Male	26	IT
	25645	3107	Sneha	Female	28	Operations
	26089	3160	Krishna	Male	38	HR
	...	...	...	...	...	...
	147046	3183	Priya	Female	26	Sales
	147232	3110	Tanvi	Female	24	Operations
	147775	3020	Sai	Male	36	Operations
	148950	3165	Simran	Female	32	Marketing
	149661	3065	Vivaan	Male	38	Finance

200 rows × 9 columns

```
In [30]: df_cleaned = df.dropna()
df_cleaned
```

```
Out[30]:
```

	Employee_ID	Employee_Name	Gender	Age	Department	Region	Exp
0	3001	Aarohi	Female	36	Sales	Assam	
1	3002	Ananya	Female	35	IT	Bihar	
2	3003	Ishita	Female	45	Finance	Uttar Pradesh	
3	3004	Vihaan	Male	56	Marketing	Uttar Pradesh	
4	3005	Dhruv	Male	31	Finance	Punjab	
...	...	...	...	...	...	...	...
195	3196	Aniket	Male	26	Finance	Uttarakhand	
196	3197	Dhruv	Male	23	HR	Odisha	
197	3198	Aryan	Male	36	Finance	Assam	
198	3199	Karthik	Male	27	Finance	Assam	
199	3200	Sneha	Female	34	HR	West Bengal	

200 rows × 10 columns

```
In [31]: df_filled = df.fillna(0)
df_filled
```

```
Out[31]:
```

	Employee_ID	Employee_Name	Gender	Age	Department	Region	Exp
0	3001	Aarohi	Female	36	Sales	Assam	
1	3002	Ananya	Female	35	IT	Bihar	
2	3003	Ishita	Female	45	Finance	Uttar Pradesh	
3	3004	Vihaan	Male	56	Marketing	Uttar Pradesh	
4	3005	Dhruv	Male	31	Finance	Punjab	
...	...	...	...	...	...	...	...
195	3196	Aniket	Male	26	Finance	Uttarakhand	
196	3197	Dhruv	Male	23	HR	Odisha	
197	3198	Aryan	Male	36	Finance	Assam	
198	3199	Karthik	Male	27	Finance	Assam	
199	3200	Sneha	Female	34	HR	West Bengal	

200 rows × 10 columns

```
In [32]: df_replaced = df.replace({'Aarohi':100})
df_replaced
```

```
Out[32]:
```

	Employee_ID	Employee_Name	Gender	Age	Department	Region	Exp
0	3001	100	Female	36	Sales	Assam	
1	3002	Ananya	Female	35	IT	Bihar	
2	3003	Ishita	Female	45	Finance	Uttar Pradesh	
3	3004	Vihaan	Male	56	Marketing	Uttar Pradesh	
4	3005	Dhruv	Male	31	Finance	Punjab	
...	...	...	...	...	...	...	...
195	3196	Aniket	Male	26	Finance	Uttarakhand	
196	3197	Dhruv	Male	23	HR	Odisha	
197	3198	Aryan	Male	36	Finance	Assam	
198	3199	Karthik	Male	27	Finance	Assam	
199	3200	Sneha	Female	34	HR	West Bengal	

200 rows × 10 columns

```
In [35]: pip install matplotlib
```

```
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: matplotlib in c:\users\hp\appdata\roaming\python\python313\site-packages (3.10.7)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\hp\appdata\roaming\python\python313\site-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cyclor>=0.10 in c:\users\hp\appdata\roaming\python\python313\site-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\hp\appdata\roaming\python\python313\site-packages (from matplotlib) (4.60.1)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\hp\appdata\roaming\python\python313\site-packages (from matplotlib) (1.4.9)
Requirement already satisfied: numpy>=1.23 in c:\users\hp\appdata\roaming\python\python313\site-packages (from matplotlib) (2.3.4)
Requirement already satisfied: packaging>=20.0 in c:\users\hp\appdata\roaming\python\python313\site-packages (from matplotlib) (25.0)
Requirement already satisfied: pillow>=8 in c:\users\hp\appdata\roaming\python\python313\site-packages (from matplotlib) (12.0.0)
Requirement already satisfied: pyparsing>=3 in c:\users\hp\appdata\roaming\python\python313\site-packages (from matplotlib) (3.2.5)
Requirement already satisfied: python-dateutil>=2.7 in c:\users\hp\appdata\roaming\python\python313\site-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in c:\users\hp\appdata\roaming\python\python313\site-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)
Note: you may need to restart the kernel to use updated packages.
```

```
In [36]: df
```

```
Out[36]:
```

	Employee_ID	Employee_Name	Gender	Age	Department	Region	Exp
0	3001	Aarohi	Female	36	Sales	Assam	
1	3002	Ananya	Female	35	IT	Bihar	
2	3003	Ishita	Female	45	Finance	Uttar Pradesh	
3	3004	Vihaan	Male	56	Marketing	Uttar Pradesh	
4	3005	Dhruv	Male	31	Finance	Punjab	
...	...	...	...	...	...	...	...
195	3196	Aniket	Male	26	Finance	Uttarakhand	
196	3197	Dhruv	Male	23	HR	Odisha	
197	3198	Aryan	Male	36	Finance	Assam	
198	3199	Karthik	Male	27	Finance	Assam	
199	3200	Sneha	Female	34	HR	West Bengal	

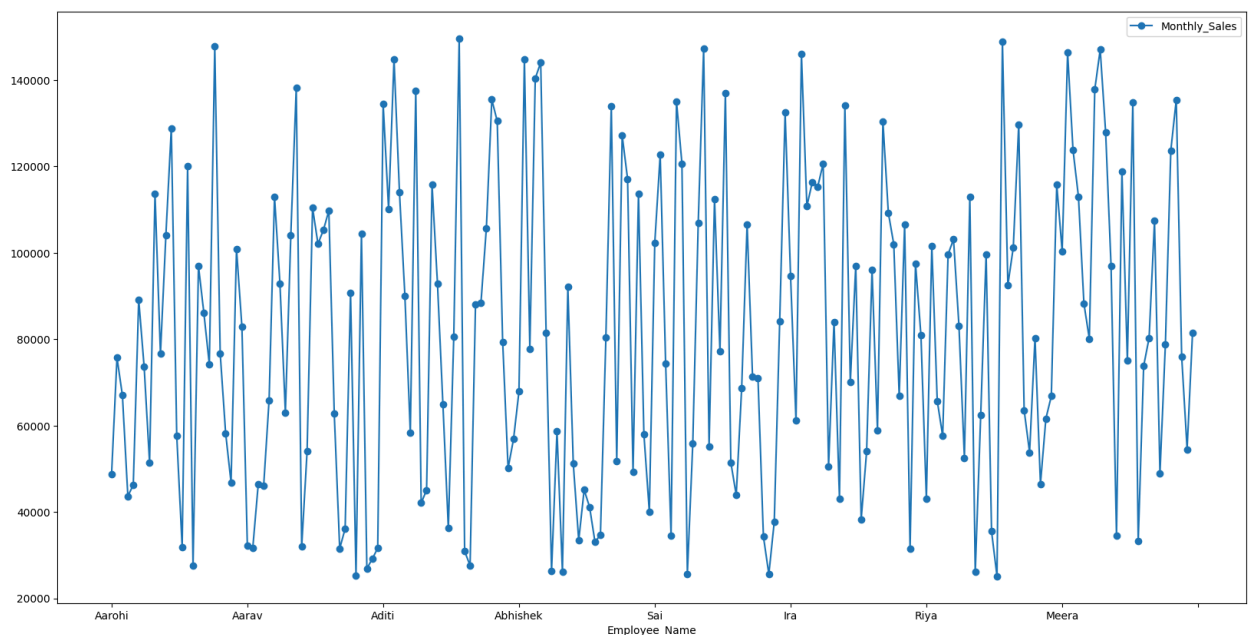
200 rows × 10 columns

```
In [ ]: import matplotlib.pyplot as plt
```

Matplotlib is building the font cache; this may take a moment.

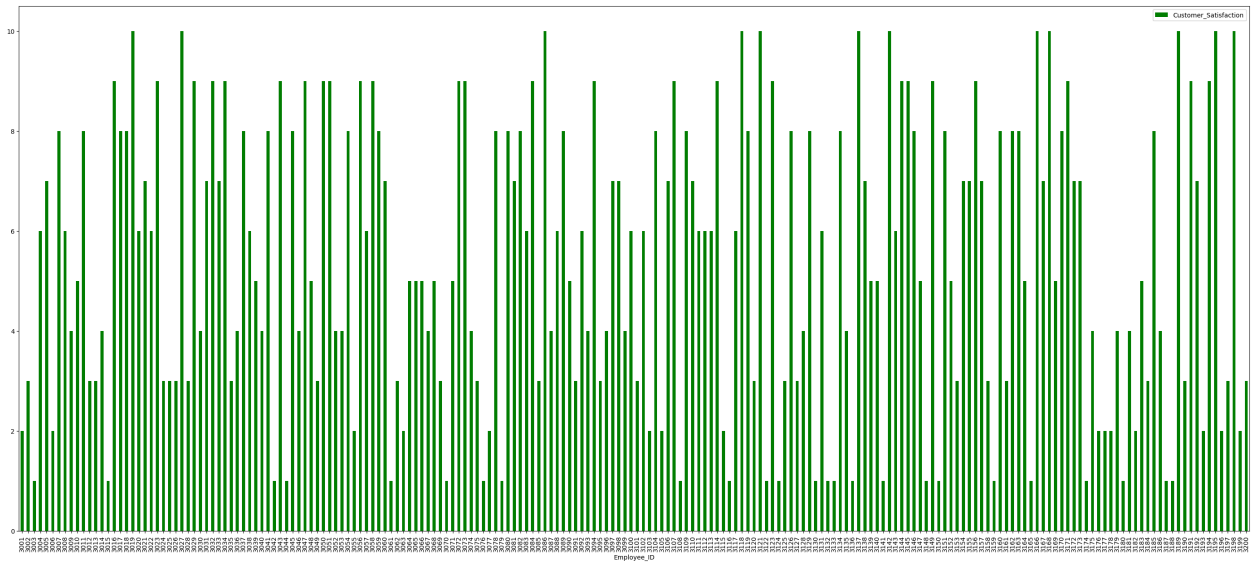
```
In [39]: df.plot(x='Employee_Name',y='Monthly_Sales',kind='line',marker='o',figsize=(20,10))
```

```
Out[39]: <Axes: xlabel='Employee_Name'>
```



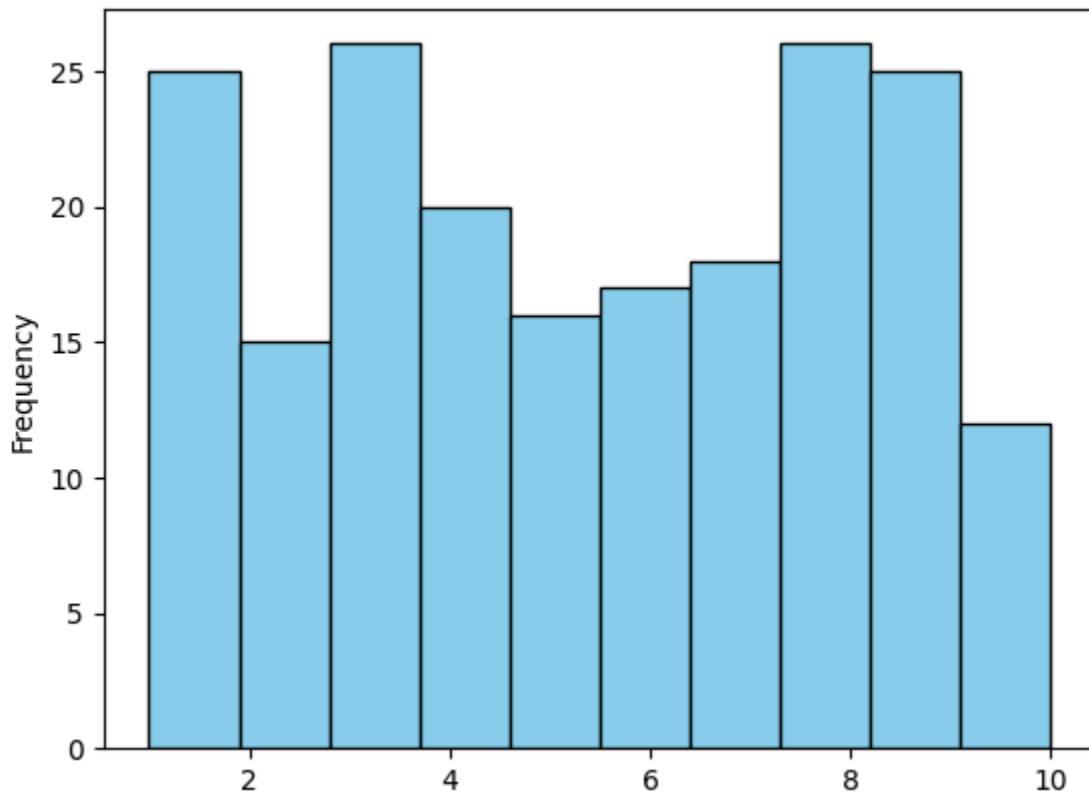
```
In [66]: df.plot(x='Employee_ID', y='Customer_Satisfaction', kind='bar', color='green',
```

```
Out[66]: <Axes: xlabel='Employee_ID'>
```



```
In [53]: df['Customer_Satisfaction'].plot(kind='hist',color='skyblue', edgecolor='black'
```

```
Out[53]: <Axes: ylabel='Frequency'>
```



```
In [67]: df.plot(x='Customer_Satisfaction', y='Employee_ID',kind='scatter',color='red',
```

```
Out[67]: <Axes: xlabel='Customer_Satisfaction', ylabel='Employee_ID'>
```

